Sale Report / Purchase Report subsection (Finance Section).

Feature Overview

- **Section:** Finance
  Sale Report (Subsection 2)
  Purchase Report (Subsection 3)

- **Scope:** Admin drawer-based view for mill-wise sales records, multi-mill synchronization, and real-time status updates synced from three distinct data modules:
  1. **Sale Soda Report**
  2. **Gate Pass (GP)**
  3. **Gate Pass Report (for Party Weight)**

Checklist & Specifications

1. UI & Navigation (Drawer & Multi-Mill Selection)

- [ ] **Drawer Component Integration**
  - Open a slide-over/drawer menu when the admin clicks on **Sale Record or Purchase Record**.
  - Add a dropdown parameter: **Select Mill**.
- [ ] **Mill-Wise Filtering**
  - Filter and display record entries strictly based on the selected mill.
- [ ] **Multi-Mill Soda Syncing Logic**
  - **Rule:** If a user selects multiple mills (e.g., 3 mills) while submitting a **Soda Form**, duplicate/reflect the same Soda entry across all 3 selected mill records.
  - Ensure unique transaction IDs maintain reference links across the synced mills.

2. Data Ingestion & Field Mapping

(Ref. table 01)

Mill Name	Goyal Agro			Sale Report																						
A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**	**AA**
Soda Date	Item	Party	Rate	Count/ Qty	Sale Order No.	Bill No.	S.No.	Date	Truck No.	GP	Bags	Bag type	Bag Weight consideration	Oil result	Oil Result after cut	Net Weight at Mill	Net Weight at Party	Lower weight	Weight Difference	Weight per bag	Rate	Bill Amount	CD%	Net Payment	Payment Status	Note
													Drop down yes or no	M												
													Backend calculati													

													on												
Mill Name	Goyal Agro			Purchase Report																					
A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**	**AA**
Soda Date	Item	Party	Rate	Count/ Qty	Purchase Order No.	Bill No.	S.No.	Date	Truck No.	GP	Bags	Bag type	Bag Weight consideration	Oil result	Oil Result after cut	Net Weight at Mill	Net Weight at Party	Lower weight	Weight Difference	Weight per bag	Rate	Bill Amount	CD%	Net Payment	Payment Status	Note
													Drop down yes or no	M											
													Backend calculation													

Source A: Sale/ Purchase Soda Report

- [ ] Automatically fetch and auto-populate the following fields upon entry creation in Sale/ Purchase Soda Report:
    - **Sold Date**
    - **Item Name**
    - **Party Name**
    - **Rate**
    - **Count / Quantity**
    - **Sale Soda Number**

Source B: Gate Pass (GP / Gate Pass Cut)

- [ ] Map and pull real-time dispatch details from the Gate Pass entry:
    - **Date**
    - **Bill No.**
    - **Truck Number**
    - **Gate Pass No.**
    - **Bags Count**
    - **Bag Type**
    - **Net Weight at Mill**

Source C: Gate Pass Report

- [ ] Fetch party-end measurement metrics:

    - **Net Weight at Party**

3. Real-Time Status Synchronization (Feedback #28 section L)

- [ ] **Payment Status Sync (Section Z)**
    - Allow Admin to toggle payment status inside **Sale Soda Record**:
        - Clear
        - Unclear
        - Underway
    - Automatically sync and display this state change in the **Soda Report with Payment status Section Z** for the respective mill(s).
    - Ensure real-time database update / websocket trigger so no page refresh is required.
    - Clearcut Payment status movement is mentioned in the (feedback 28)

4. Database Requirements

- [ ] **DB Schema / Relationships:** Ensure Sale Report maps correctly to Sale Soda Report, GatePass, and Gate Pass Report foreign keys.
- [ ] **Validation:** Verify weight discrepancies between *Net Weight at Mill* and *Net Weight at Party*.
- [ ] **Audit Trail:** Log status changes made by admins on payment statuses.

Feature Overview

- **Scope:** Real-time data pipeline propagating entries from **Soda Forms** through **Gate Pass (GP) dispatches** into **Sale/Purchase Reports** and **Soda Reports**.
- **Key Mechanics:**
    4. Multi-mill distribution from a single Soda Form.
    5. Real-time entry creation upon GP dispatch.
    6. Chronological cross-mill serial numbering (**Section H**).
    7. Strict filtering to ignore saved draft states.

Developer Checklist & Specifications

1. Soda Form Creation & Initialization

- [ ] **Multi-Mill Allocation**
    - When a Soda Form is created for multiple mills (e.g., 3 Rice Bran trucks split across Beas Agro, Goyal Agro, and Rajoa Agro), reflect the master Soda record across all designated mills.
- [ ] **Draft Handling & Validation (Crucial)**
    - **Rule:** If a GP selecting this Soda Number is saved as a **Draft**, DO NOT push, sync, or record any entry in the **Soda Report**, **Sale Report**, or **Purchase Report**.
    - **Rule:** Data sync triggers **ONLY** upon final GP **Submission / GP Out / In Dispatch**.

2. Gate Pass (GP) Linkage & Real-Time Sync

- [ ] **Real-Time Data Pipeline Trigger**
  - Upon hitting **Submit/Dispatch (GP Out/ GP In)**:
    - Immediately reflect the dispatched transaction in the **Sale Soda Report / Purchase Soda Report**.
    - Immediately push a line item into the **Sales Report** (or **Purchase Report**, depending on transaction type) of the respective mill executing the dispatch. In front of the respective soda No. (depending on transaction type).

3. Chronological Serial Numbering Logic (Section H - Table 1)

- [ ] **Global Chronological Sequencing**
  - Implement global timestamp tracking based on the exact moment the GP is cut/submitted (created_at / dispatched_at timestamp).
- [ ] **Cross-Mill Sequential Assignment (Section H)**
  - Assign incremental serial numbers (01, 02, 03, etc.) to each truck dispatch belonging to the **same parent Soda Number**, regardless of which mill dispatches it.

Case Study Verification Workflow:

```
Master Soda Entry: 3 Rice Bran Trucks Created

1st GP Dispatch (10:00 AM) @ Beas Agro  ---> Beas Sales Report  :
Serial # 01
2nd GP Dispatch (11:30 AM) @ Goyal Agro ---> Goyal Sales Report :
Serial # 02
3rd GP Dispatch (02:00 PM) @ Rajoa Agro ---> Rajoa Sales Report :
Serial # 03
```

- [ ] **Section H Column Integration**
  - Map this calculated sequential integer directly into **Section H (Table 1)** of the respective mill's Sales/Purchase Report.

4. Database & Logic Rules

- [ ] **Race Condition Safeguard:** Use database transactions/locks when generating **Section H** serial numbers to ensure two trucks dispatched simultaneously from different mills do not receive identical serial numbers.
- [ ] **Audit Trail:** Retain parent-child relationships (Soda_ID \rightarrow GP_ID \rightarrow Report_Row_ID) for tracking and verification.

Bag Weight Consideration (Section N) logic, including conditional dropdown handling, standard weight constants, and automated deductions from net weight fields.

Feature Overview

- **Section:** Finance > Sale/Purchase Report > Table 1 (Section N)
- **Scope:** Dynamic toggle allowing admins to deduct standard tare weight of packaging bags (Plastic vs. Jute) from the recorded gross weights at both the Mill (Section Q) and the Party (Section R).

Developer Checklist & Specifications

1. UI Component & UI Control (Section N)

- [ ] **Bag Weight Consideration Dropdown**
  - Add a required dropdown field in **Section N** of Table 1.
  - Options: Yes | No
- [ ] **Default State:** Set default selection to No unless explicitly toggled by the admin.

2. System Constants & Conversion Standard

Define and lock the standard bag tare weight constants in the backend configuration file to maintain high numeric precision:

- [ ] **Plastic Bag Standard Weight:** 0.002 { quintals/bag} (200 grams)
- [ ] **Jute Bag Standard Weight:** 0.010 { quintals/bag} (1 kg)

3. Calculation Logic & Dynamic Deduction

Case 1: Admin Selects No

- [ ] Skip bag weight deduction entirely.
- [ ] Pass original weights directly: {Displayed Net Weight (Section Q/R)} = {Unadjusted Net Weight}

Case 2: Admin Selects Yes

- [ ] Execute real-time backend/frontend calculation using the formula:
  Total Bag Weight (Quintals) = {Bags Count} * {Bag Weight Constant}
- [ ] Subtract *Total Tare Weight* from both ***Section Q*** *and* ***Section R***:
  Adjusted Net Weight at Mill (Section Q) = Net Weight at Mill - Total Bag Weight
  Adjusted Net Weight at Party (Section R) = Net Weight at Mill - Total Bag Weight

4. Execution Examples for Verification

- [ ] **Test Case 1: Plastic Bags**
  - **Input:** 500 bags (from Section L), Bag Type = Plastic (from Section M), Section N = Yes
  - **Calculation:** 500 * 0.002 = 1.00{ quintal}
  - **Result:** Deduct 1.00 { quintal} from Section Q (Net Weight at Mill) and Deduct 1.00 { quintal} Section R (Net Weight at Party).
- [ ] **Test Case 2: Jute Bags**
  - **Input:** 500 bags (from Section L), Bag Type = Jute (from Section M), Section N = Yes
  - **Calculation:** 500 * 0.010 = 5.00{ quintals}
  - **Result:** Deduct 5.00{ quintals} from Section Q (Net Weight at Mill) and Deduct 5.00{ quintals} Section R (Net Weight at Party).

5. Guardrails & Edge Cases

- [ ] **Dependency Check:** Section N calculation must listen to **Bags Count (Section L)** and **Bag Type (Section M)** fetched from the Gate Pass. If either value changes, recalculate Section Q and Section R instantly.
- [ ] **Negative Weight Prevention:** Throw an error if {Total Bags Weight} > {Net Weight}.
- [ ] **Floating Point Precision:** Ensure standard decimal rounding (up to 3 decimal places for quintals) to avoid floating-point rounding errors in report calculations.

Oil Result Section (Section O) and **Oil Result After Cut (Section P)** logic.

Feature Overview

- **Section:** Finance > Sale/Purchase Report > Table 1 (Sections O & P)
- **Scope:** Manual input field for lab-tested Rice Bran Oil percentage result, paired with an automated backend calculation for standard percentage deduction ("cut").

Developer Checklist & Specifications

1. UI & Input Field (Section O: Oil Result)

- [ ] **Manual Admin Input Field**
  - Provide a numerical input field in **Section O** of Table 1 labeled **"Oil Result"**.
  - Restrict inputs strictly to positive numeric/decimal values (e.g., 18.5, 19.0).
- [ ] **Validation Guardrails**
  - Display inline validation if an admin enters non-numeric text or a value \le 0.
  - Require entry in Section O for Rice Bran items to calculate Section P.

2. Backend Calculation Logic (Section P: Oil Result After Cut)

- [ ] **Automated Deductive Formula**
  - Calculate **Section P** automatically whenever **Section O** is entered or updated:

    A = Oil Result (Section O) - 1

    Then the **If** formula runs

    =If(A>15,(A-15)/2+15,A)

- [ ] **Read-Only Field Handling**
  - Make **Section P** a non-editable / read-only calculated field in the UI.

3. Developer Verification Examples

- [ ] **Test Case 1**
  - **Input (Section O):** 18.5%
  - **Calculation A :** 18.5 - 1 = 17.5%
  - **Formula Evaluation: = If(17.5>15,(17.5-15)/2+15,17.5)**
  - Here the condition of 17.5>15 stands true, so further calculation will happen
  - (17.5-15)/2+15=16.25 will appear
  - **Output (Section P):** 16.25%
- [ ] **Test Case 2**
  - **Input (Section O):** 14.0%
  - **Calculation:** 14.0 - 1 = 13.0%
  - **Formula Evaluation: = If(14>15,(14-15)/2+14,14)**
  - Here the condition 14>15 stand false, so as per the formula the figure will remain same
  - **Output (Section P):** 19.0%

4. System Integration & Triggers

- [ ] **Item-Type Dependency:** Enable Sections O and P specifically when the item category is **Rice Bran** (or items requiring lab oil percentage records).
- [ ] **Real-Time UI Update:** Ensure Section P updates dynamically on keypress/input change in Section O without requiring a full page save.

Section S (Lower Weight), **Section T (Weight Difference)**, and **Section U (Weight Per Bag)** of Table 1.

Feature Overview

- **Section:** Finance > Sale/Purchase Report > Table 1 (**Sections S, T, and U**)
- **Scope:** Dynamic calculation section handling billing weight determinations, cross-location weight variance tracking (Mill vs. Party), and per-unit bag weight metrics.

Developer Checklist & Specifications

1. Section S: Lower Weight (Billing Weight)

- [ ] **Business Purpose:** Determines the official billable net weight by taking the minimum value between Mill and Party weights.
- [ ] **Input Dependencies:**
  - **Section Q:** Net Weight at Mill
  - **Section R:** Net Weight at Party
- [ ] **Backend / Frontend Calculation Logic:**
  Section S (Lower Weight) = min(Section Q, Section R)
- [ ] **Display & Billing Integration:**
  - Highlight this field visually in the UI as **Primary Billing Weight**.
  - Map this value directly as the reference weight for invoice/billing payload generation.

2. Section T: Weight Difference (Mill vs. Party Variance)

- [ ] **Business Purpose:** Measures exact weight discrepancies between origin (Mill) and destination (Party) to track shortage or excess during transit.
- [ ] **Input Dependencies:** Section Q and Section R.
- [ ] **Calculation Logic:**
  Section T (Weight Difference) = Section Q (Mill Weight) - Section R (Party Weight)
- [ ] **Rules & Formatting:**
  - **Allow Negative Figures:** Do not apply absolute values (Math.abs). Allow signed values explicitly:
    - **Positive Value (+):** Indicates Mill weight was higher
    - **Negative Value (-):** Indicates Party weight was higher
  - **UI Visual Indicator:** Display + or - clearly (e.g., +0.50 Quintals or -0.25 Quintals).

3. Section U: Weight Per Bag

- [ ] **Business Purpose:** Calculates the average net weight per bag based on Mill weighment records.
- [ ] **Input Dependencies:**
    - **Section Q:** Net Weight at Mill
    - **Section L:** Bags Count (fetched from Gate Pass)

- [ ] **Calculation Logic:**
  Section U (Weight Per Bag) = Section Q / Section L

- [ ] **Validation & Zero-Division Safeguard:**
    - **Zero Check:** If Section L = 0 or null, return 0 or N/A to prevent division-by-zero application crashes.
    - **Precision:** Format output to standard 3 decimal places (e.g., 0.502 Quintals/Bag)

4. Verification & Test Cases

Test Field	Section Q (Mill)	Section R (Party)	Section L (Bags)	Expected Section S (Lower Wt)	Expected Section T (Diff)	Expected Section U (Wt/Bag)
Case 1	500.00	495.00	1,000	**495.00**	**+5.00**	**0.500**
Case 2	480.00	485.00	800	**480.00**	**-5.00**	**0.600**
Case 3	500.00	500.00	1,000	**500.00**	**0.00**	**0.500**

Logic updates in both Sale and Purchase reports:

Developer Logic Implementation Checklist

- [ ] **Section V: Rate Calculation Logic**
    1. [ ] **Standard Logic:** Bind Section V Rate directly to the Section D Rate figure (values update simultaneously).

    2. [ ] **Special Case (Rice Bran Selected):**
        - [ ] Detect when item/product selected is "Rice Bran".
        - [ ] Calculate Section V Rate using formula:
          Section V Rate =Oil Result after Cut (Section P) * Rate (Section D)

- [ ] **Section W: Bill Amount Calculation**
    - [ ] Calculate Section W Bill Amount using formula:
      Bill Amount(Section W) = Rate (Section V) * Lower Weight(Section S)
      [ ] Ensure rules for Section V and Section W apply uniformly across both **Sale Report** and **Purchase Report**.
- [ ] **Section X: Cash Discount (CD) Percentage**
    - [ ] Provide a manual input field for Admin to enter custom percentage values (e.g.,

1%, 1.5%, etc.).

- [ ] **Section Y: Net Payment Calculation & Routing**
  - [ ] Calculate Net Payment using backend formula:
    CD Amount = Bill Amount (Section W) * CD% (Section X)
    Net Payment (Section Y) = Bill Amount (Section W) - CD Amount

  - [ ] **Soda Sheet Integration:** Display Section Y Net Payment in the sum total of the corresponding Soda Number on the respective Soda Sheet.

  - [ ] **Ledger Integration (Data Hook):** Map Section Y Net Payment to mirror into the party's respective Ledger Account to drive Accounts Payable / Accounts Receivable (detailed logic deferred to next feedback).

- [ ] **Section Z: Payment Status**
  - [ ] Maintain implementation as specified in Page 3 feedback.

- [ ] **Section AA: Note**
  - [ ] This will be Manual input.